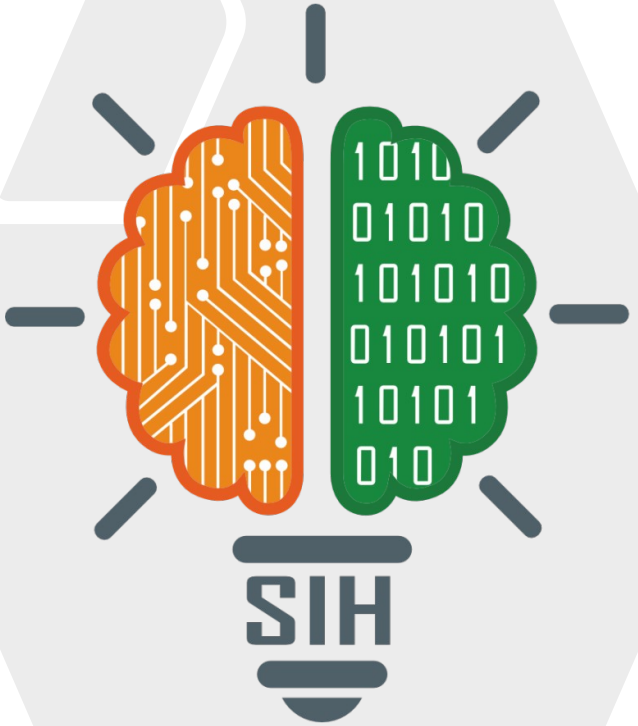
# SMART INDIA HACKATHON 2026

# SCOURGE

*Ask the bank, not the paper.*




- **Problem Statement ID –** SIH26265
- **Problem Statement Title –** Streamline of Security Deposit / FDR Portal
- **Theme –** Smart Automation
- **PS Category –** Software
- **Team ID –** 199322
- **Team Name (Registered on portal) –** Arnavax#

SCOURGE

# SCOURGE — ASK THE BANK, NOT THE PAPER

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## Proposed Solution (Describe your Idea/Solution/Prototype)

**An institution proves what its bank's own server says about one FDR. AICTE sees only that record.**

### Detailed explanation of the proposed solution

AICTE sends a **single-use challenge** for one FDR. The institution logs into its own net banking inside one TLS session whose keys are **split with AICTE's verifier** (MPC-TLS). Password, OTP, PAN and other accounts never leave the institution.

### How it addresses the problem

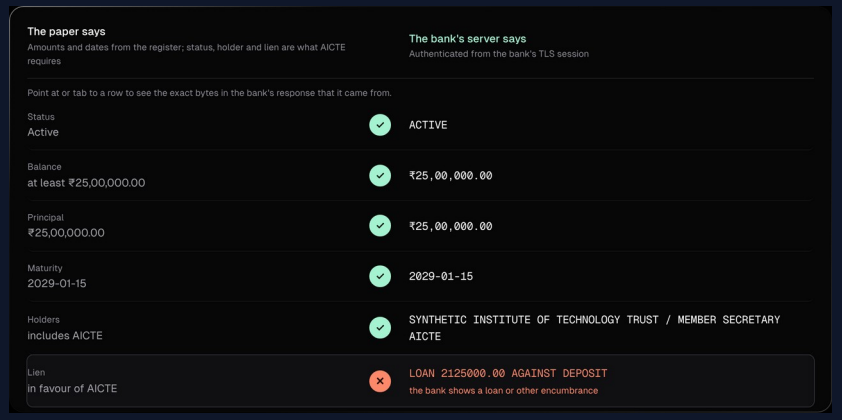
Paper is checked once and trusted for 10 years. Scourge **re-checks the bank any time and before every release**: a loan, early encashment, missing pledge (lien) or fake FDR shows up as a **Discrepancy**.

### Innovation and uniqueness of the solution

**Works on every bank's existing net-banking page, and API-ready**. Three evidence tiers: bank-session proof, then bank-signed e-advice, then branch confirmation. Split key: the institution cannot edit what the bank said; fakes and replays are rejected.

LIVE PROTOTYPE · SYNTHETIC BANK

**The bank's server contradicts the paper.**




× HIDDEN LOAN CAUGHT

Every value traces to the bank's own bytes

## The complete FDR lifecycle in one portal (all five stages run in the prototype)

**1 FILE**
Institution files online; renewals linked

→

**2 PROVE**
Bank-session proof on AICTE's challenge

→

**3 MONITOR**
T-90/60/30/7 alerts; adequacy; MT940/camt.053 reconcile

→

**4 DECIDE**
Verdict + evidence; hash-chained audit

→

**5 PLEDGE RELEASE**
Dept clearances, 2 signers, holds; QR-signed letter

**Why now:** **₹201.42 crore of FDRs "do not appear in IDFC's records"** (CBI chargesheet, Oct 2026); AICTE joint FDRs found encashed or loaned against without consent.

SCOURGE

# TECHNICAL APPROACH

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## Technologies to be used

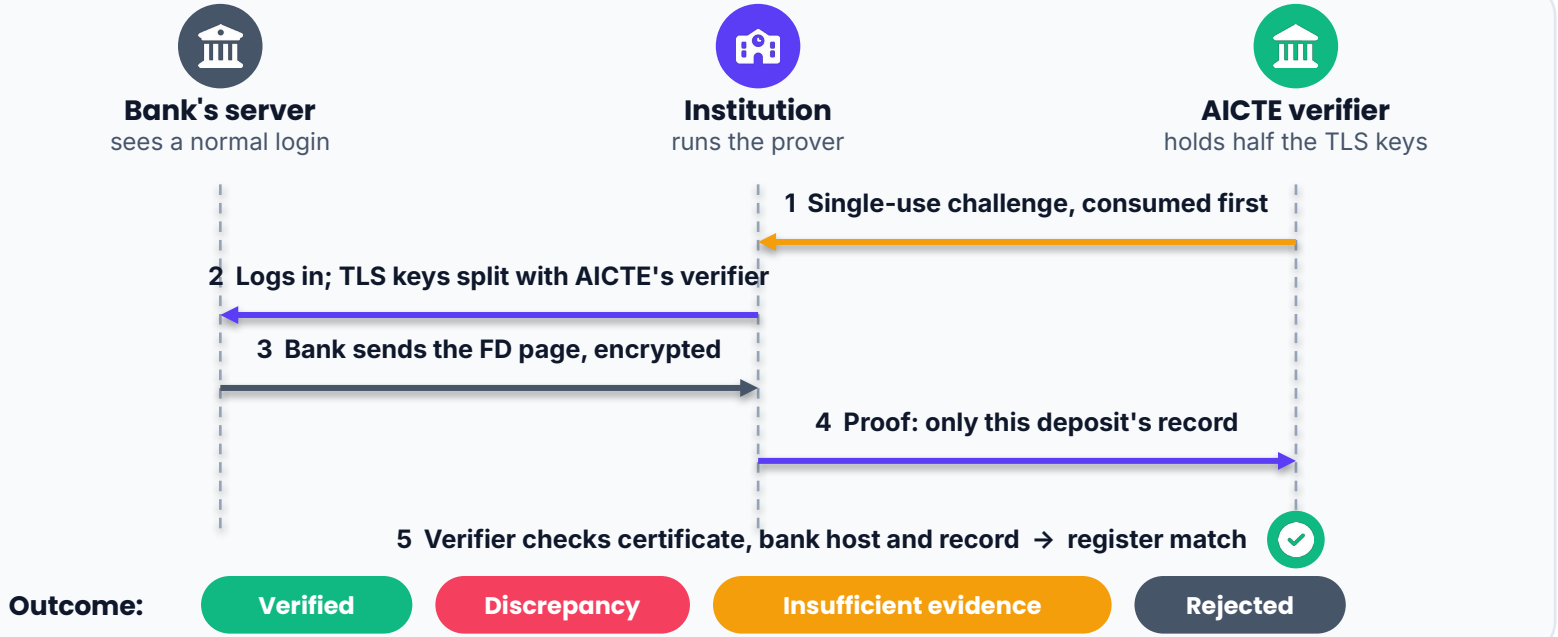
**Proof layer**
TLSNotary MPC-TLS · Rust
Selective disclosure · rustls

**Backend**
Python · FastAPI · SQLite
AES-GCM vault · hash-chained audit

**Frontend**
Server-rendered HTML · JS
Live 3-party proof view

**Ops & QA**
Docker · nginx · GitHub CI
181 tests · 2FA · audits

## Methodology and process for implementation (Flow Charts/Images/ working prototype)




**WATCH THE 90-SECOND DEMO**




**Click to play here, or scan / click the QR for YouTube**
Real proofs on a synthetic bank

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# FEASIBILITY AND VIABILITY

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## Analysis of the feasibility of the idea

**0.58 s**
**per bank proof**
*median of 60 runs · synthetic bank*

**SBI live**
**live split-key session, SBI API**
*HTTP 200 in 1.0 s · 30 Oct 2026*

**181 / 181**
**automated tests pass**
*incl. attack, forgery, 2FA, vault tests*

**₹0**
**software licences**
*one small server [estimate]*

- **Technical:** Open-source TLSNotary (MIT/Apache-2.0); runs on one small server
- **Operational:** Institution uses its own login; no bank integration needed
- **Viability:** Adopted by an AICTE circular; covers the existing FDR book

## Potential challenges and risks

## Strategies for overcoming these challenges

| | |
|---|---|
| **Bank page hides liens, or bytes are withheld** | Fallbacks: bank-signed e-advice, then branch confirmation; never a guess |
| **Fake bank site or a replayed old proof** | Rejected: bank certificate + single-use, 15-min challenge |
| **Proof of a different FD, or an edited screen** | Rejected / no effect: FD bound to challenge; proof covers bank bytes |
| **Bank needs browser-bound session / TLS 1.3** | Browser-extension prover; bank marked unsupported until then |

**ROADMAP** 1 FD-page proof at SBI (live SBI session already done) → 2 Standard FDR-pledge schema for all banks → 3 Pilot with an AICTE regional office → 4 Bank confirmation API as banks onboard; e-BG for new deposits

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# IMPACT AND BENEFITS

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## Potential impact on the target audience

### AICTE RB officers
A dated proof from the bank in seconds, not a letter in weeks

### AICTE finance
The whole FDR book is re-checkable, before every release

### Institutions
One short proof from their own net banking; Hindi or English

### Students
The deposit that protects them is checked across all 10 years

## Benefits of the solution (social, economic, environmental, etc.)

| TODAY | WITH SCOURGE |
|---|---|
| Branch letter or uploaded screenshot | **The bank's own server, proven** |
| Checked once, trusted for 10 years | **Re-checked any time, before every release** |
| Weeks per confirmation | **Under a second per proof (synthetic bank)** |
| Hidden loans found at release, or never | **Pledge checked at the bank; two-officer release** |

**TESTED ON 180 FRESH DEPOSITS**
Random amounts, dates, formats · real proofs · synthetic bank

| | |
|---|---|
| **119 / 119** | bad deposits the upload portal approved |
| **0 / 119** | bad deposits Scourge verified |
| **0 / 61** | honest deposits Scourge accused |

**Social**
students' deposits stay intact

**Economic**
stops loss before release

**Environmental**
no paper letters or couriers

**Governance**
2FA, encrypted vault, chained audit

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# RESEARCH AND REFERENCES

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## Details / Links of the reference and research work

### The problem is real

CBI: ₹201.42 cr of FDRs not in IDFC's records (ThePrint, 7 Oct 2026)
theprint.in › cbi-says-haryana-pollution-board...

CBI on AICTE: joint FDRs encashed or loaned without consent
deccanherald.com › greater-transparency-aicte...

### Regulation

AICTE guidelines 2024-27: cl. 6.2(g) 10-year release; cl. 11.8
indiankanoon.org/doc/173919027

Account Aggregator users must be regulated FIUs (DFS)
financialservices.gov.in › account-aggregator...

### Technology

TLSNotary: MPC-TLS proofs of web data, MIT / Apache-2.0
tlsnotary.org

NeSL e-BG for security deposits: new instruments only (WDRA)
wdra.gov.in › ebg_requirement-27-08-2025

**PROJECT RESOURCES**

**Demo video (90 s):** youtu.be/8HoplhFOGCg
**Prototype:** Scourge v1.0.0-rc1 · 181 / 181 tests · synthetic bank
**Code + evidence:** github.com/Lakki007/Arnavax-Scourge-SIH26265

## Scourge vs existing approaches

| | Upload & approve | Bank letter | Account Aggregator | NeSL e-BG | Scourge |
|---|---|---|---|---|---|
| Uses the bank's own data | ✗ | ✓ | ✓ | ✓ | ✓ |
| No bank API or agreement | ✓ | ✓ | ✗ | ✗ | ✓ |
| Institution cannot forge it | ✗ | ~ | ✓ | ✓ | ✓ |
| Covers existing FDR book | ✓ | ✓ | ✓ | ✗ | ✓ |
| Re-check in seconds | ✗ | ✗ | ✓ | ✓ | ✓ |
| Shown on a real bank today | ✓ | ✓ | ✓ | ✓ | SBI session; FD next |

*~ a bank letter can still be forged. Live split-key session with SBI's API done (30 Oct 2026); FD-page proof is roadmap step 1.*